

Derivation of a Quantile Confidence Interval Based on Order Statistics

Frederik Danielsen

This note provides a mathematical derivation of a confidence interval for a quantile ξ_q based on order statistics. The results presented here are classical and are not original to this note. They are derived independently to ensure a thorough understanding of the underlying mathematics, to present the arguments in a form suited to the Problab¹ project, and to obtain conclusions that are directly useful for the implementation in Problab. The derivation assumes i.i.d. samples $X_1, \dots, X_n$ satisfying $P(X_i \leq \xi_q) = q$ and a significance level $\alpha \in (0, 1)$, corresponding to a target coverage level of $1 - \alpha$. In the first lemma, a confidence interval of the form $[X_{(k_L^*)}, X_{(k_U^*+1)}]$ is derived by choosing the indices k_L^* and k_U^* such that the two binomial tail probabilities are bounded by $\alpha/2$. In the second lemma, it is shown that the iterative procedures used in the implementation terminate at exactly these optimal indices.

Practically, these results enable Problab to return finite-sample confidence intervals with coverage probability at least $1 - \alpha$ for estimated quantiles, such as medians or percentiles.

Lemma 1. *Let X_i with $i \in \{1, \dots, n-1\}$ be i.i.d samples. Suppose that $P(X_i \leq \xi_q) = q$ for all i , some $q \in (0, 1)$, and ξ_q in the value space of X_i . Let*

$$I_i = \begin{cases} 1, & X_i \leq \xi_q, \\ 0, & X_i > \xi_q. \end{cases}$$

and let

$$K = \sum_{i=1}^n I_i \sim \text{Bin}(n, q).$$

Let k_L^* and k_U^* solve

$$\begin{aligned} k_L^* &= \max_{k_L \in \{1, \dots, n-1\}} \left\{ k_L : P(K < k_L) \leq \frac{\alpha}{2} \right\} \\ k_U^* &= \min_{k_U \in \{1, \dots, n-1\}} \left\{ k_U : P(K > k_U) \leq \frac{\alpha}{2} \right\}. \end{aligned}$$

Then

$$[X_{(k_L^*)}, X_{(k_U^*+1)}]$$

is a confidence interval for ξ_q with coverage probability at least $1 - \alpha$. Furthermore, among the intervals satisfying the two tail constraints, solving the optimization problems above produces the narrowest interval.

¹<https://github.com/FrederikDanielsenDevelopment/problab>

Proof. I_i can be regarded as the outcomes of individual Bernouli trials with probability of success q with

$$K = \sum_{i=1}^n I_i \sim \text{Bin}(n, q)$$

denoting the number of samples with values at most ξ_q . Now, consider the ordered samples

$$X_{(1)}, X_{(2)}, \dots, X_{(n)}.$$

If $K \leq k_U$ for some $k_U \in \{1, \dots, n-1\}$, that would imply that at most k_U samples fell below ξ_q . That is, $\xi_q < X_{(k_U+1)}$. Likewise, if $K \geq k_L$ for some $k_L \in \{1, \dots, n-1\}$, that would imply that at least k_L samples fell below ξ_q . That is, $\xi_q \geq X_{(k_L)}$. Hence,

$$P(k_L \leq K \leq k_U) = P(X_{(k_L)} \leq \xi_q < X_{(k_U+1)}).$$

Therefore, if we determine k_L and k_U such that

$$P(k_L \leq K \leq k_U) = 1 - \alpha,$$

for some $\alpha \in (0, 1)$, then the interval $[X_{(k_L)}, X_{(k_U+1)}]$ is a confidence interval for ξ_q with coverage probability $1 - \alpha$. However, this equality is one with two unknowns and thus multiple solutions. However, we may impose an additional constraint, namely that the likelihood that ξ_q lies above the confidence interval is the same as the likelihood that it lies below it – at least to the extent that a discrete number of samples allows it. Specifically, we require that

$$P(\xi_q \geq X_{(k_U+1)}) = P(\xi_q < X_{(k_L)}) = \alpha/2.$$

However, as mentioned, since we are dealing with a discrete number of samples, it is likely that no (k_L, k_U) exists that fulfil this exactly. Therefore, we make do with the best estimate (likely more conservative than $1 - \alpha$) and require simply that

$$P(\xi_q \geq X_{(k_U+1)}) \leq \alpha/2,$$

$$P(\xi_q < X_{(k_L)}) \leq \alpha/2$$

$$\Leftrightarrow$$

$$P(K > k_U) \leq \alpha/2,$$

$$P(K < k_L) \leq \alpha/2.$$

Since $\alpha < 1$, any pair (k_L, k_U) satisfying both tail constraints necessarily satisfies $k_L \leq k_U$. Indeed, if $k_L > k_U$, then every possible value of K satisfies either $K < k_L$ or $K > k_U$, and therefore

$$1 = P((K < k_L) \cup (K > k_U)) \leq P(K < k_L) + P(K > k_U) \leq \alpha,$$

which is a contradiction. Furthermore, in case you were wondering, since the lower endpoint $X_{(k_L)}$ requires $k_L \geq 1$ and the upper endpoint $X_{(k_U+1)}$ requires $k_U \leq n-1$, the inequality $k_L \leq k_U$ implies that any feasible pair must satisfy

$$k_L, k_U \in \{1, \dots, n-1\}.$$

To obtain the narrowest confidence interval satisfying these constraints, we choose the largest possible k_L and the smallest possible k_U . Thus, we solve

$$\begin{aligned} k_L^* &= \max_{k_L \in \{1, \dots, n-1\}} \left\{ k_L : P(K < k_L) \leq \frac{\alpha}{2} \right\}, \\ k_U^* &= \min_{k_U \in \{1, \dots, n-1\}} \left\{ k_U : P(K > k_U) \leq \frac{\alpha}{2} \right\}. \end{aligned}$$

In this way we ensure that

$$P(k_L^* \leq K \leq k_U^*) = P(X_{(k_L^*)} \leq \xi_q < X_{(k_U^*+1)}) \geq 1 - \alpha.$$

Then the interval

$$[X_{(k_L^*)}, X_{(k_U^*+1)}]$$

is at least as conservative as a confidence interval with coverage probability $1 - \alpha$. $\square$

Lemma 2. *Let $K \sim \text{Bin}(n, q)$ and $\alpha \in (0, 1)$, and suppose that the optimization problems*

$$\begin{aligned} k_L^* &= \max_{k \in \{1, \dots, n-1\}} \left\{ k : P(K < k) \leq \frac{\alpha}{2} \right\}, \\ k_U^* &= \min_{k \in \{1, \dots, n-1\}} \left\{ k : P(K > k) \leq \frac{\alpha}{2} \right\} \end{aligned}$$

have solutions. Consider the iterative procedures

$$\begin{aligned} k_L &\leftarrow 1, \\ \text{while } k_L < n-1 \text{ and } P(K \leq k_L) &\leq \frac{\alpha}{2}, \quad k_L \leftarrow k_L + 1, \end{aligned}$$

and

$$\begin{aligned} k_U &\leftarrow n-1, \\ \text{while } k_U > 1 \text{ and } P(K \geq k_U) &\leq \frac{\alpha}{2}, \quad k_U \leftarrow k_U - 1. \end{aligned}$$

Then the procedures terminate with

$$k_L = k_L^* \quad \text{and} \quad k_U = k_U^*.$$

Proof. For the lower index, observe that increasing the current value k_L to $k_L + 1$ is permissible exactly when

$$P(K < k_L + 1) = P(K \leq k_L) \leq \frac{\alpha}{2}.$$

Thus, at every iteration, the algorithm increments k_L only when the next larger value satisfies the lower-tail constraint. Since k_L increases by one at each iteration and is bounded above by $n - 1$, the procedure terminates. If the procedure stops before reaching $n - 1$, then increasing k_L by one would violate the constraint. Indeed, the next candidate would be $k_L + 1$, and for this candidate the constraint is

$$P(K < k_L + 1) \leq \frac{\alpha}{2}.$$

Since

$$P(K < k_L + 1) = P(K \leq k_L),$$

and the loop has stopped, this inequality is no longer satisfied. Therefore, $k_L + 1$ is not feasible. Since $P(K < k)$ is nondecreasing as k increases, no larger value can be feasible either. Hence, the returned k_L is the largest feasible value, so

$$k_L = \max_{k \in \{1, \dots, n-1\}} \left\{ k : P(K < k) \leq \frac{\alpha}{2} \right\} = k_L^*.$$

If the procedure instead terminates at $k_L = n - 1$, then $n - 1$ is itself the largest possible value in the domain, so again $k_L = k_L^*$. For the upper index, observe that decreasing the current value k_U to $k_U - 1$ is permissible exactly when

$$P(K > k_U - 1) = P(K \geq k_U) \leq \frac{\alpha}{2}.$$

Thus, at every iteration, the algorithm decrements k_U only when the next smaller value satisfies the upper-tail constraint. Since k_U decreases by one at each iteration and is bounded below by 1, the procedure terminates. If the procedure stops before reaching 1, then decreasing k_U by one would violate the constraint. Indeed, the next candidate would be $k_U - 1$, and for this candidate the constraint is

$$P(K > k_U - 1) \leq \frac{\alpha}{2}.$$

Since

$$P(K > k_U - 1) = P(K \geq k_U),$$

and the loop has stopped, this inequality is no longer satisfied. Therefore, $k_U - 1$ is not feasible. Since $P(K > k)$ increases as k decreases, no smaller value can be feasible either. Hence, the returned k_U is the smallest feasible value, so

$$k_U = \min_{k \in \{1, \dots, n-1\}} \left\{ k : P(K > k) \leq \frac{\alpha}{2} \right\} = k_U^*.$$

If the procedure instead terminates at $k_U = 1$, then 1 is itself the smallest possible value in the domain, so again $k_U = k_U^*$. $\square$